

Factoring Notes

What factoring is, how to do it, and how the SAT uses it

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1 What is factoring?

Factoring means rewriting an expression as a **product**: pieces that are multiplied together. It is the reverse of distributing (or FOIL).

You already do this with numbers. Factoring 12 means writing it as $3 \cdot 4$ or $2 \cdot 2 \cdot 3$. With expressions, it looks like this:

TWO DIRECTIONS

$$(x + 2)(x + 3) \text{ — distribute / FOIL } \rightarrow x^2 + 5x + 6$$

$$x^2 + 5x + 6 \text{ — factor } \rightarrow (x + 2)(x + 3)$$

Why bother?

Once an expression is written as a product, you can do things you couldn't before:

- **Solve equations.** If $(x - 5)(x + 2) = 0$, then $x = 5$ or $x = -2$.
- **Simplify fractions.** Cancel a factor that appears on the top and the bottom.
- **Find zeros and x-intercepts** of a function just by reading the factors.
- **Find unknown constants** by matching a factored form to a standard form.

Words to know

Word	Meaning	Example
Term	Pieces that are added or subtracted	$x^2 + 5x + 6$ has 3 terms
Factor	Pieces that are multiplied	$(x + 2)$ and $(x + 3)$
Coefficient	The number in front of a variable	the 5 in $5x$
Constant	A number with no variable	the 6 in $x^2 + 5x + 6$
GCF	Greatest common factor: the biggest thing every term shares	the GCF of $6x^2$ and $9x$ is $3x$
Factor completely	Keep going until no factor can be factored again	$2x^2 - 18 = 2(x + 3)(x - 3)$

2 Step one, always: the GCF

Before anything else, check whether every term shares something. If it does, pull it out front.

1. Find the biggest number that divides every coefficient.
2. For each variable that appears in every term, take the **smallest** power.
3. Write the GCF outside the parentheses, and divide each term by it to get what goes inside.

EXAMPLE: GCF

$$12x^3 + 18x^2$$

Numbers: the biggest number dividing 12 and 18 is 6. Variable: smallest power is x^2 . GCF = $6x^2$.

$$12x^3 + 18x^2 = 6x^2(2x + 3)$$

$$\text{Check: } 6x^2 \cdot 2x = 12x^3 \text{ and } 6x^2 \cdot 3 = 18x^2 \checkmark$$

The common factor can be a whole chunk

Sometimes the thing every term shares is a whole binomial in parentheses. Treat it exactly like a single variable. This is the move behind **Problem 3**:

EXAMPLE: A common binomial (Problem 3)

$$[6x(6x + 10) + 4(6x + 10)] / (3x + 5)$$

Both terms on top contain $(6x + 10)$. Pull it out: top = $(6x + 4)(6x + 10)$.

Now pull a 2 out of $(6x + 10)$: it becomes $2(3x + 5)$, which matches the bottom.

$$(6x + 4) \cdot 2(3x + 5) / (3x + 5) = 2(6x + 4) = 12x + 8$$

The same idea works with absolute values. That's **Problem 7**:

EXAMPLE: A repeated absolute value (Problem 7)

$$8|x - 3| - 11|x - 3| = -48$$

Think of $|x - 3|$ as one thing, like y . Then $8y - 11y = -3y$, so $-3|x - 3| = -48$.

Divide by -3 : $|x - 3| = 16$.

So $x - 3 = 16$ or $x - 3 = -16$, which gives $x = 19$ or $x = -13$.

TIP

If you see the same chunk more than once, pretend it's a single letter. Factor or combine, then put the chunk back.

3 Trinomials that start with x^2

For $x^2 + bx + c$, look for **two numbers that multiply to c and add to b** . Those two numbers go in the parentheses.

EXAMPLE: $x^2 + 7x + 12$

Pairs that multiply to 12: $1 \cdot 12$, $2 \cdot 6$, $3 \cdot 4$

Which pair adds to 7? $3 + 4 = 7$ ✓

$$x^2 + 7x + 12 = (x + 3)(x + 4)$$

Sign rules

The signs of b and c tell you the signs in the answer before you start:

If c is...	and b is...	then the signs are...	Example
positive	positive	both +	$x^2 + 7x + 12 = (x + 3)(x + 4)$
positive	negative	both -	$x^2 - 7x + 12 = (x - 3)(x - 4)$
negative	positive	one + and one -; the bigger number is +	$x^2 + x - 12 = (x + 4)(x - 3)$
negative	negative	one + and one -; the bigger number is -	$x^2 - x - 12 = (x - 4)(x + 3)$

4 Trinomials that don't start with x^2

For $ax^2 + bx + c$ where a isn't 1, use the **AC method**:

1. Multiply $a \cdot c$.
2. Find two numbers that multiply to $a \cdot c$ and add to b .
3. Use those two numbers to split the middle term into two terms.
4. Factor by grouping (Section 6).

EXAMPLE: $2x^2 + 7x + 3$

$a \cdot c = 2 \cdot 3 = 6$. Two numbers that multiply to 6 and add to 7: 1 and 6.

Split: $2x^2 + 1x + 6x + 3$

Group: $x(2x + 1) + 3(2x + 1)$

Answer: $(2x + 1)(x + 3)$

EXAMPLE: $3x^2 - 10x - 8$

$a \cdot c = 3 \cdot (-8) = -24$. Multiply to -24 and add to -10: -12 and 2.

Split: $3x^2 - 12x + 2x - 8$

Group: $3x(x - 4) + 2(x - 4)$

Answer: $(3x + 2)(x - 4)$

SAT SHORTCUT

On multiple-choice questions you can skip the factoring: FOIL each answer choice and see which one gives back the original.

5 Special patterns

Difference of squares

$$a^2 - b^2 = (a + b)(a - b)$$

Use it when there are **two terms**, both are **perfect squares**, and there's a **minus** between them.

EXAMPLE: Difference of squares

$$x^2 - 49 = (x + 7)(x - 7)$$

$$9x^2 - 25 = (3x + 5)(3x - 5) \text{ because } 9x^2 = (3x)^2 \text{ and } 25 = 5^2$$

WATCH OUT

A **sum** of squares like $x^2 + 49$ does **not** factor. Only the minus version works.

Perfect square trinomials

$$a^2 + 2ab + b^2 = (a + b)^2$$

$$a^2 - 2ab + b^2 = (a - b)^2$$

Spot it when the first and last terms are perfect squares and the middle term is 2 times their square roots multiplied.

EXAMPLE: Perfect square

$$x^2 + 10x + 25: x^2 = (x)^2, 25 = 5^2, \text{ and } 2 \cdot x \cdot 5 = 10x \checkmark$$

$$\text{So } x^2 + 10x + 25 = (x + 5)^2$$

SAT SHORTCUT

If $a^2 - b^2 = 48$ and $a - b = 4$, you don't need a or b .

Write $(a + b)(a - b) = 48$, so $(a + b)(4) = 48$ and $a + b = 12$.

6 Factoring by grouping

Use this when there are **four terms**.

1. Split the four terms into two pairs.
2. Pull the GCF out of each pair.
3. If the two parentheses match, pull that matching chunk out.

EXAMPLE: Grouping

$$x^3 + 3x^2 + 4x + 12$$

$$\text{Pairs: } (x^3 + 3x^2) + (4x + 12)$$

$$\text{GCF of each: } x^2(x + 3) + 4(x + 3)$$

$$\text{Matching chunk } (x + 3): \text{ answer } (x^2 + 4)(x + 3)$$

WATCH OUT

When the second pair starts with a minus, pull out a **negative** GCF so the parentheses match:

$$x^3 + 2x^2 - 5x - 10 = x^2(x + 2) - 5(x + 2) = (x^2 - 5)(x + 2)$$

7 How the SAT uses factoring

A. Solving equations: the zero product property

If two things multiply to 0, at least one of them must be 0. So once you factor, set each factor equal to 0.

EXAMPLE: Solving

$$x^2 - 3x = 10$$

$$\text{First move everything to one side: } x^2 - 3x - 10 = 0$$

$$\text{Factor: } (x - 5)(x + 2) = 0$$

$$\text{So } x - 5 = 0 \text{ or } x + 2 = 0, \text{ which gives } x = 5 \text{ or } x = -2.$$

WATCH OUT

The equation must equal **0** before you factor. $(x - 5)(x + 2) = 8$ does **not** mean $x - 5 = 8$.

B. Zeros and x-intercepts

A factor $(x - r)$ means the function equals 0 at $x = r$. For $g(x) = (x - 8)(x - 7)(x - 4)$ the zeros are 8, 7, and 4. For $h(x) = g(x + 3)$ (Problem 4), set the inside equal to each zero: $x + 3 = 8, 7, 4$, so $x = 5, 4, 1$.

C. Simplifying fractions

Factor the top and the bottom completely, then cancel factors that match.

EXAMPLE: Cancelling

$$(x^2 - 9) / (x^2 + 5x + 6) = (x - 3)(x + 3) / ((x + 2)(x + 3)) = (x - 3) / (x + 2)$$

WATCH OUT

You can only cancel **factors** (things being multiplied), never **terms** (things being added).

$(x^2 - 9) / (x^2 + 5x + 6)$ is **not** $-9 / (5x + 6)$. The x^2 's can't be crossed out.

D. Matching a factored form

When an equation says a standard form equals a factored form, FOIL the factored side and match the terms one at a time. This is the idea behind **Problem 11**.

EXAMPLE: Matching

$$x^2 + kx + 18 = (x + 3)(x + m)$$

FOIL the right side: $x^2 + (3 + m)x + 3m$

Match constants: $3m = 18$, so $m = 6$.

Match the x terms: $k = 3 + m = 9$.

8 Which method? A checklist

Step	Ask yourself	If yes...
1	Do all the terms share something?	Pull out the GCF first (Section 2).
2	Two terms, both squares, with a minus?	Difference of squares (Section 5).
3	Three terms starting with x^2 ?	Find two numbers that multiply to c and add to b (Section 3).
4	Three terms starting with a number times x^2 ?	AC method (Section 4). Check for a perfect square too.
5	Four terms?	Grouping (Section 6).
6	Can any factor be factored again?	Keep going until nothing more factors.
7	Done?	Check by multiplying back out or plugging in a number (Section 10).

9 Common mistakes

Mistake	Wrong	Right
Dropping the GCF	$2x^2 - 18 = (x + 3)(x - 3)$	$2x^2 - 18 = 2(x + 3)(x - 3)$
Factoring a sum of squares	$x^2 + 9 = (x + 3)(x - 3)$	$x^2 + 9$ doesn't factor
Sign error in grouping	$x^2(x + 2) - 5(x - 2)$	$x^2(x + 2) - 5(x + 2)$
Cancelling terms	$(x + 6) / 6 = x$	$(x + 6) / 6$ doesn't simplify
Not setting = 0 first	$x(x - 3) = 10 \rightarrow x = 10$	$x^2 - 3x - 10 = 0 \rightarrow x = 5 \text{ or } -2$
Stopping too early	$x^4 - 16 = (x^2 + 4)(x^2 - 4)$	$(x^2 + 4)(x + 2)(x - 2)$

10 Check your answer

Method 1: Multiply it back out

FOIL or distribute your answer. You should get the original expression exactly.

Method 2: Plug in an easy number

Pick a number like $x = 2$ (avoid 0 and 1, since they hide mistakes). Put it into the original and into your answer. If the results match, you're almost certainly right. On multiple choice, plug the same number into every choice and keep the one that matches.

EXAMPLE: Plugging in on Problem 3

Original at $x = 2$: $[12(22) + 4(22)] / 11 = 352 / 11 = 32$

Choice D at $x = 2$: $12(2) + 8 = 32 \checkmark$